CSE 422 Homework I

1) Theorem: If A_1 and A_2 are regular languages, so is $A_1 \cap A_2$.

Proof: Let $M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$ recognizes A_1.

$M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$ recognizes A_2.

Construct $M = (Q, \Sigma, \delta, q_0, F)$ recognizing $A_1 \cap A_2$

M should accept input w if both M_1 and M_2 accept it.

Components of M:

$Q = Q_1 \times Q_2$

$q_0 = (q_1, q_2)$

$\delta((q_1, r), a) = (\delta_1(q_1, a), \delta_2(q_2, a))$

$F = F_1 \times F_2$

A language is regular if some finite automaton recognizes it.

2)

a) $M = (Q, \Sigma, \delta, q_0, F)$ where:

Q: Finite set of states = {a, b, c, d, e, f}

Σ: Finite set of alphabet symbols = {0, 1}

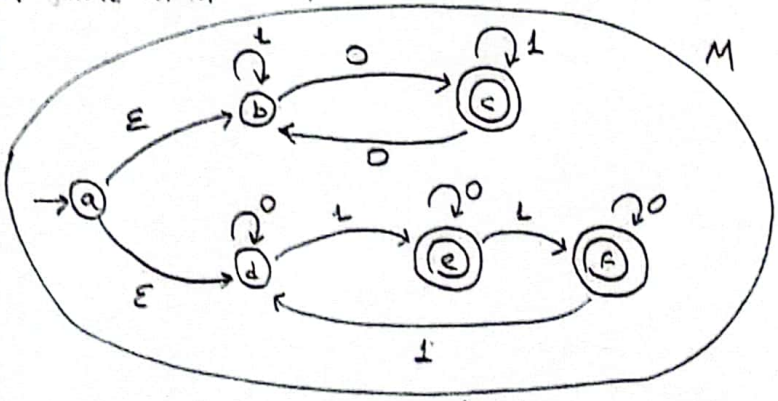
δ: Transition functions = $Q \times \Sigma_\varepsilon \to P(Q)$ ↳ Power set of Q

q_0: Start state = a

F: finite set of accept states = {c, e, f}

$L(M)$ = {w | w has an odd number of 0's or the number of 1's in w is not a multiple of 3.}

b)



c)

Yes, it is possible.

$L(M) = (1^*0(0^*10^*1)^*)\cup((0^*10^*)\cup(0^*10^*1(0^*\cup(10^*10^*1))^*))$

6) No, M may recognize other language as well such as subsets of A language.

d) Input: 01100

Transition	From	To	With
1	a	b	ε
2	b	c	0
3	c	c	1
4	c	c	1
5	c	b	0
6	b	(c)	0

All input consumed.

* C is accept state and no input is left.

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3)

a) The rules follow the form of CFG for both of them.

For the first one:

CFG = (V, Σ, R, S) where

- V = {S, A}
- Σ = {a, b}
- R is the rules.
- S is S.

For the second one:

CFG = (V, Σ, R, S) where

- V = {S, T}
- Σ = {a, b}
- R is the rules.
- S is S.

b) The first CFG is not ambigous because there is only one way to parse the string. However the second CFG is ambigous because there are different ways of parsing it.

For the string "ab":

1st way: $S \to aSb \to aTb \to ab$ ($S \to T$) ($T \to \varepsilon$)

2nd way: $S \to aTb \to ab$

c) Description of Pushdown Automata:

1) Place the start variable on the stack.

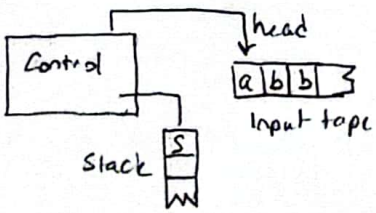
2) Repeat the following steps forever.

a) If the top of stack is variable symbol, nondeterministically select one of the rules for it and substitute it by right hand side of the rule.

b) If the top of stack is terminal symbol, read the symbol from input tape and compare them. If they match, repeat. Otherwise, reject on this branch of nondeterminism.

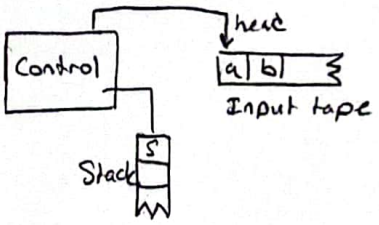
c) If the stack is empty, enter the accept state.

View for first PDA:



Stack	Input
S	abb
aSb	abb
Sb	bb
Abb	bb
εbb	bb
b	b
b	b

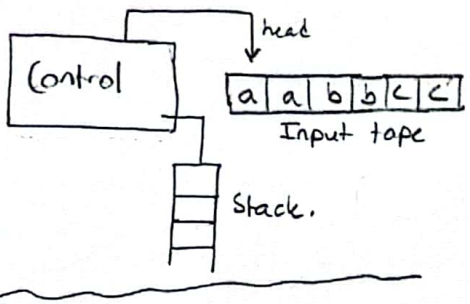
View for second PDA



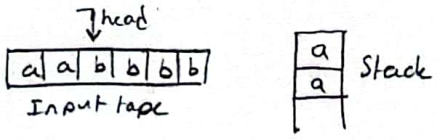
Stack	Input
S	ab
aSb	ab
Sb	b
εb	b

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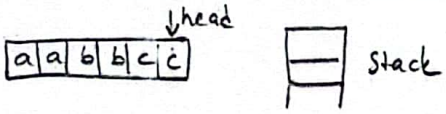
4) P



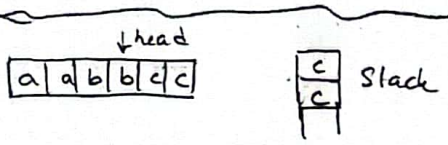
It reads to the right until a's end. Also, it pushes 'a' to the stack for each 'a'.

① 

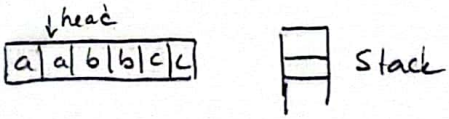
Then, it reads to the right until b's end. Also, it pop 'a' from the stack for each 'b'. Finally, head goes to the end of the input tape.

② 

Now, stack is empty. P reads to the left until C's end. Also, it pushes 'c' to the stack for each 'c'.

③ 

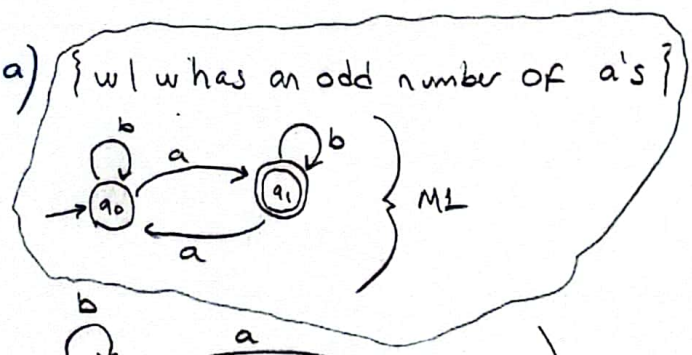
Then it reads to the left until b's end. Also, it pop 'c' from the stack for each 'b'.

④ 

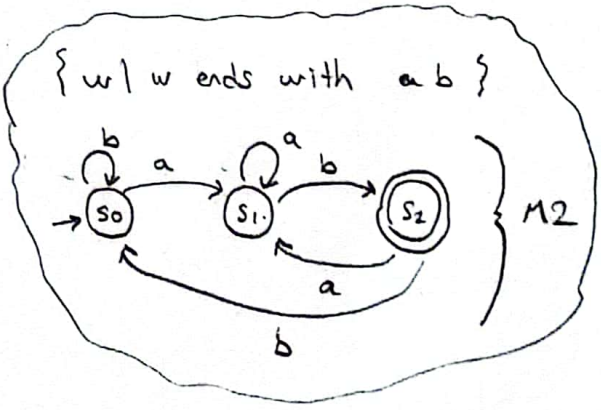
Finally, all input is checked and stack is empty. So, the PDA recognizes the string "aabbcc" for the language $\{a^i b^i c^i \mid i \geq 0\}$

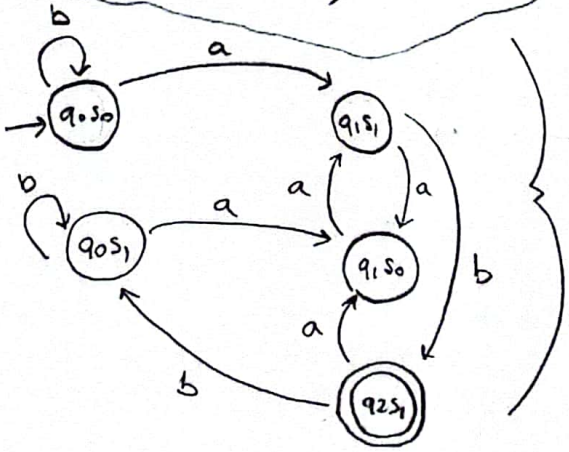
5)

a) $\{w \mid w \text{ has an odd number of a's}\}$



$\{w \mid w \text{ ends with a b}\}$

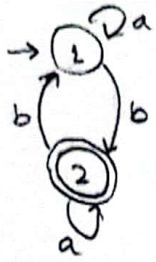




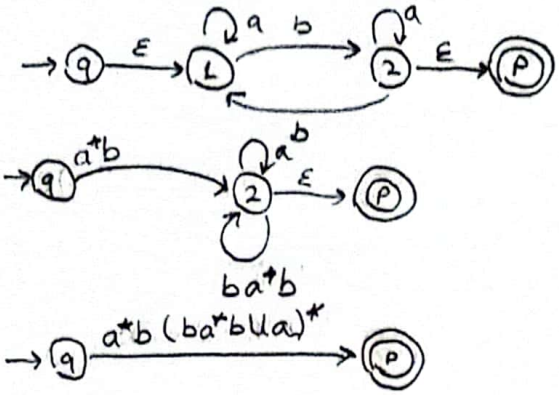
$M_1 \cap M_2$

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b)



First converting it to GNFA.



d) The language F is the union of three disjoint languages.

$F_0 = \{b^j c^k \mid j,k \geq 0\}$, $F_1 \{a b^j c^j \mid j \geq 0\}$, $F_2 = \{a^i b^j c^k \mid i \geq j, k \geq 0\}$

Clearly F_0 and F_2 are regular languages.
The class of regular languages are closed under union, intersection and complement.
Thus $\overline{F_0 \cup F_2}$ is also regular.

We have $F_1 = F \cap \overline{F_0 \cup F_2}$

So, if F is regular, so is F_1.

a) Assume that F_1 is regular language and p is the pumping length.

Consider a string $ab^p c^p = S = uvw$ such that $|uv| \leq p$, $|v| > 0$ and $uv^i w \in F_1$ $\forall i \geq 0$.

Take $u = a$, $v = b^p$, $w = c^p$

$uv^0 w = a(b^p)^0 c^p = ac^p \notin F_1 \rightarrow$ It breaks the structure of the language.

So, F_1 is not regular language, so is F.

b) Let p is 2.

Consider a string $a^i b^j c^k \in F$ of at least 2.

$x = \epsilon$

- if $i \neq 2$, y is first symbol of $a^i b^j c^k$
- if $i = 2$, $y = aa$

Clearly, it satisfies the pumping lemma conditions.

c) Part a and b do not contradict the pumping lemma because it says if a language is regular then there is a pumping length for that language. But, if the language satisfies pumping lemma, the language may not be regular.

e)

c) $001 \cup 0^* 1^*$

The string 001 is in language but if it is generated by 001 then it cannot be pumped. If string s is larger than 3 and in the language, then it is generated by $0^* 1^*$. Dividing the string according to Pumping Lemma's condition into a string xyz where x is ϵ y be the first symbol and z be the remaining, user can pump the string. So, the minimum pump length is 4.

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h) $10(11^*0)^*0$

The minimum length string in the given language is 100 but cannot be pumped. Next minimum length string is 10100. If divide it according to the pumping lemma in xyz then x be 10, y be 10 and z be the rest of the string then y can be pumped. Hence minimum pumping length is 4.

j) Σ^*

Say s be a string in the language Σ^*. According to pumping lemma if divide s in xyz then x be the empty string, y is $(\varepsilon|0|1)$ and z is empty string. Now ε couldn't be pumped. So, the minimum pumping length is 1.

f) $\{w \mid w \in w^R$, that is, w is a palindrome$\}$

$S \rightarrow 0$

$S \rightarrow 1$

$S \rightarrow 0S0$

$S \rightarrow 1S1$

$S \rightarrow \varepsilon$

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